Speech Recognition (DSAI 456)

Lecture 1: What is Speech System?

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Agenda

Objectives: you should be able to say what a speech system takes in and what it emits, compute the error rate of one by hand, and explain why a leaderboard number does not tell you whether the system works for your users.

- Problem: What does the machine actually receive, and why is turning it into text hard?
- Measurement: What is WER, what does it hide, and when is it the wrong metric?
- Course: How is this course run, and what does the project ask of you?

Core Speech Processing Tasks

Key functional modalities in modern speech AI systems.

ASR

Automatic Speech Recognition

Converts continuous acoustic audio signals into discrete text transcripts.

`Audio → Text`

TTS

Text-to-Speech

Synthesizes human-like spoken audio waveforms directly from input text.

`Text → Audio`

Speech LM

Speech Language Models

Generates speech directly from speech by preserving tone and emotion.

`Audio → Audio`

What the Machine Actually Receives

RAW INPUT STREAM

16,000 samples/sec

Continuous amplitude values (at 16 kHz)

TARGET OUTPUT

12–15 chars/sec

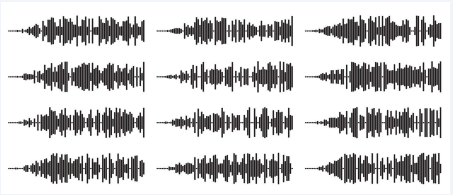
Discrete text characters per second

SCALE DISPARITY

~1,000 : 1

Ratio of acoustic data points to output text

Continuous Acoustic Waveform



No boundaries between words or phonemes

The Core Challenge: Nothing in the physical wave signal indicates where one word ends and the next begins.

You cannot look at a waveform and see the word. Speech sound is stretched out over time and mixed together with background noise, room echoes, and the speaker's voice.

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- **The input and output run at different rates.** The model has to decide which stretch of audio produced which character, and nobody hands it that alignment.
- **Ambiguity is only resolved by context.** Identical acoustics, different text, and only the surrounding words decide. A perfect acoustic model still needs a language model.

Why Not Just Classify?

Classification

- One input, one label from a fixed set. Ten thousand X-rays, fourteen diagnoses.
- The output space is known before training. You can count the classes.
- The question is which one is this?
- Success means matching the label.

Sequence transduction

- One input of arbitrary length, one output sequence of a different and unknown length.
- The output space is every string the language admits. You cannot enumerate it.
- The question is what sequence explains this signal?
- Success means a transcript close to what was said, and "close" needs defining.

Word Error Rate

Align the hypothesis to the reference with edit distance, count the three kinds of mistake, divide by the length of the **reference**.

$$\mathrm{WER} = \frac{S + D + I}{N}$$

Reference	the committee **approved** the **new** budget **today**
Hypothesis	the committee **improved** the budget **to date**
	2 substitution, 1 deletion, 1 insertion

$$\mathrm{WER} = 4/7 = 57\%$$

Important
WER has no upper bound. If you ever report a WER above 1.0, that is not a bug.

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- **The average hides the tail.** A 7% WER over a corpus can be 3% for the men reading news and 30% for one dialect speaker. Your project must report per-speaker and per-condition breakdowns.

What To Take Away

- Speech recognition maps a long, continuous, hugely variable signal to a short discrete sequence, with no alignment given and no boundaries marked.
- WER is edit distance normalized by reference length. It has no ceiling, it weights every error equally, and it changes when your text normalizer changes.
- A leaderboard number describes a corpus, not a language and not your users. Ask what was spoken, by whom, and what the model had already seen.

Next week: the acoustics. Formants, sampling, quantization, framing, F0 and intensity.

Before We Meet Again

1. **Browse the Open ASR Leaderboard.** Pick the top system and write down, in one sentence each, which corpus its headline number comes from and which population that corpus represents.

2. **Record 30 seconds of yourself** on your phone, in Arabic, speaking the way you speak to a friend rather than to a microphone. Run it through any free ASR system. Compute the WER by hand.

3. **Bring one failure.** One error the system made that a human listener would not have made, and one sentence on why you think it happened.

> **Tip**
> Start thinking about teams now. The proposal is due in week 4, and the teams that go looking for a real problem early are the ones that end up with something worth demoing.

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